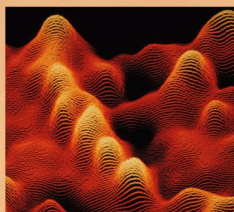


DNA

KNOWN AS THE "MOLECULE OF LIFE", THE CHEMICAL DNA (DEOXYRIBONUCLEIC ACID) CONTAINS THE INSTRUCTIONS, KNOWN AS GENES, FOR THE BODY'S GROWTH, FUNCTION, AND REPAIR.

In nearly all human cells, DNA is packaged into 46 coiled structures called chromosomes, situated in the cell's nucleus. DNA's list of instructions takes the form of long, thin molecules, one per chromosome, each forming a double-helix shape. Each double-helix has two long strands that corkscrew around each other. These are linked by rungs, like a ladder. The rungs are made of pairs of chemicals called bases: adenine (A), guanine (G), thymine (T), and cytosine (C). A always pairs with T, and G with C. The order of the bases contains the chromosome's genetic code, while the way the bases link enables DNA to make copies of itself.



DNA UNDER THE MICROSCOPE

This scanning tunnelling micrograph (STM) of DNA, magnified about one million times, shows the twists of the helix as a series of yellow peaks on the left.

BASE PAIRS

The four bases can pair in only two configurations due to their chemical structures. Adenine and thymine each have two positions for forming hydrogen bonds and so fit together, while guanine and cytosine each have three hydrogen-bond locations.

